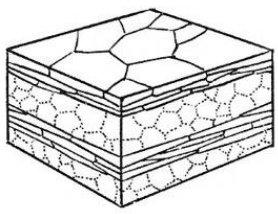
**EART11300 practical: Igneous, sedimentary and metamorphic rocks in hand specimen - answers**

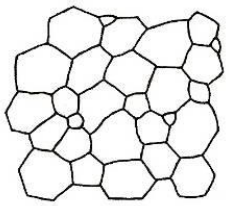
| Rock | Made of clasts? yes/no | see inter-locking crystals? y/n | Crystals have own shape? y/n | Compos-itional layering? y/n | Foliation ? y/n | Fossils or organic matter? y/n | Glassy? y/n | Pores or vesicles? y/n | Ig/Met/Sed |
|---|---|---|---|---|---|---|---|---|---|
| **L106** | no | no | no | yes | no | yes | no | no | sedimentary |
| **R5** | no | no | yes | no | no | no | no | yes | igneous |
| **R14** | no | yes | yes | no | no | no | no | no | igneous |
| **R17** | no | no | yes | no | no | no | no | no | igneous |
| **R26** | yes | no | no | yes | no | no | no | yes | sedimentary |
| **R31** | yes | no | no | no | no | yes | no | yes | sedimentary |
| **R45 or A1** | no | no | no | yes | yes | no | no | no | metamorphic |
| **R47** | no | maybe | no | no | no | no | no | no | metamorphic |
| **S77** | yes | no | no | no | no | no | no | no | sedimentary |
| **S110** | no | no | Too small to tell | no | no | yes | no | no | sedimentary |
| **W12** | no | yes | yes | no | no | no | no | no | igneous |
| **X** | no | no | Too small to tell | no | no | no | no | yes | igneous |
| **Z** | no | no | Only the red ones (garnet) | no | no | no | no | no | metamorphic |
| **Φ1 or Φ2** | no | no | Too small to tell | A bit in Φ1 | yes | no | no | no | metamorphic |
| **Φ4 or Φ5** | no | no | Some of them | yes | yes | no | no | no | metamorphic |

**Minerals you may be able to identify, and rock names (that you are not expected to know):**

L106: limestone (contains fossil stromatolites)
R5: olivine basalt (the green crystals are the mineral olivine)
R14: quartz + feldspar + biotite. granite.
R17: quartz (the glassy-looking crystals). The white crystals are feldspar. rhyolite.
R26: sandstone
R31: calcite. limestone (contains fossil crinoid stems)
R45 or A1: quartz + feldspar + biotite. gneiss. Contains both layering and foliation (see sketch on right).
R47: calcite. marble.
S77: conglomerate
S110: mudstone



W12: gabbro [contains feldspar (white) + pyroxene + olivine].
X: basalt. It is too sparkly to be a sedimentary rock, in which crystals would be weathered and dull-looking. The white blobs are vesicles in which minerals have precipitated from aqueous solution after the rock had solidified – these are called amygdales.
Z: garnet amphibolite (you may be able to recognise garnet from last week). Only the garnet has its own shape; the other minerals have shapes like soap bubbles (see sketch on right).



Φ1 & Φ2: slate. The fact that the rock has split into slabs with smooth surfaces, indicates that there is a foliation.
Φ4 & Φ5: mica (muscovite + biotite) + quartz. You may also be able to recognise garnet from last week. garnet mica schist.

## C. Further classification of the different rock types:

**Igneous Rocks.**
A simple classification of igneous rocks is based upon the grain size, which tells us how quickly the rock has cooled:

- Medium- to coarse-grained = slow cooling = **plutonic**
- Fine-grained = fast cooling = **volcanic**

Note that if the rock contains larger crystals in a finer-grained "groundmass" (= the rock matrix), then this indicates a 2-stage cooling process, with the larger crystals growing slowly in a magma chamger, and the groundmass crystallising in the eruption.
In the following table, indicate whether the igneous rocks are plutonic or volcanic.

| Plutonic | Volcanic |
|---|---|
| R14 | R17 |
| W12 | X |
| | R5 |

**Sedimentary Rocks.**
If the rock is made of particles, then it can be classified according to grain size, which indicates the distance travelled from source to deposition:

- **Proximal** = close to source = coarser grain size.
- **Distal** = far away from source = finer grain size.

The grain size reflects the amount of energy available for transportation. This classification does not work so well for rocks containing fossils whose size is determined by the kind of fossil rather than distance travelled.
In the following table, arrange your sedimentary samples into 3 categories from closest to source (coarsest grain size) to furthest away from source.

| | Sample numbers |
|---|---|
| Closest to source | S77, R31 |
| | R26, L106 |
| Furthest from source | S110 |

**Metamorphic Rocks.**
Can be classified according to the extent that the rock has changed during metamorphism, which usually depends on the temperature reached, and is known as **metamorphic grade.**

- **Low grade** display the least amount of change. If the rock started out as fine-grained (e.g. a mudstone), then low-grade metamorphism will result in a rock that is still fine-grained.
- **Medium grade** show an intermediate level of change. It will be coarser grained.
- **High grade** show the most change. It will have the largest crystals.

In the following table, arrange your metamorphic samples into 3 categories from lowest grade (finest grain size) to highest grade. Note that it can be difficult to compare the metamorphic grade of rocks with very different compositions.

| | Sample numbers |
|---|---|
| Lowest grade | Φ1 & Φ2 |
| | R47, Φ4 & Φ5 |
| Highest grade | R45, Z |